Offline Facial Recognition & Liveness Detection System for Remote Locations

Problem Statement

Field personnel working in remote and zero-network zones face challenges in secure identity verification. Existing authentication systems depend heavily on internet connectivity, making them unreliable in remote environments. There is a need for a lightweight, secure, and completely offline authentication system that can accurately verify personnel identity and prevent spoofing attacks.

Proposed Solution

We propose an AI-powered mobile application that performs offline facial recognition and liveness detection on Android and iOS devices. The system verifies user identity locally without requiring internet access and supports synchronization with cloud infrastructure when connectivity is restored.

Key Features

- Offline facial recognition
- Offline liveness detection using blink, smile, and head movement verification
- React Native cross-platform application
- Local encrypted storage
- AWS synchronization support
- Lightweight AI model optimized for mobile devices
- Fast authentication process (<1 second)

Technology Stack

- React Native
- TensorFlow Lite
- MobileFaceNet
- SQLite
- AWS Cloud Services
- Open-source AI frameworks

System Workflow

1. User Login
2. Face Capture
3. Liveness Verification
4. Facial Recognition
5. Authentication Decision
6. Local Storage
7. Cloud Synchronization when network becomes available

Expected Impact

The proposed solution enables secure, reliable, and efficient personnel authentication in remote environments while maintaining privacy, performance, and operational continuity.

Future Scope

- Advanced anti-spoofing mechanisms
- Multi-factor authentication
- Improved edge AI optimization
- Real-time analytics dashboard
- Large-scale deployment support